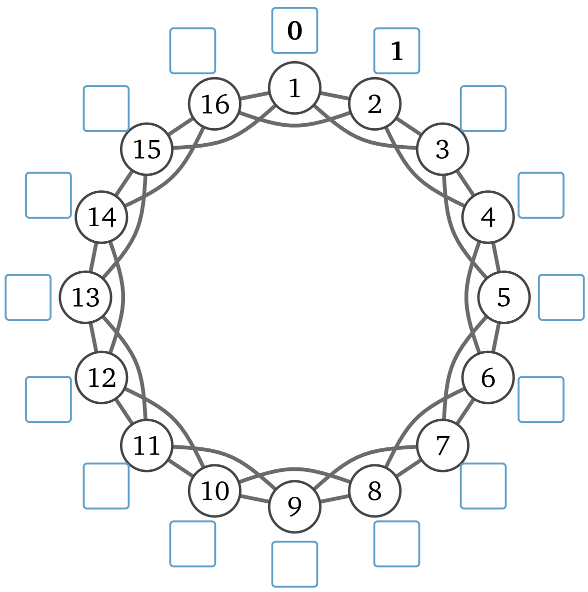
Part 1 — A letter from Omaha

In 1967 a man in Omaha, Nebraska was handed a letter addressed to a stockbroker outside Boston — a stranger, thirteen hundred miles away. He could not post it. He could pass it only to somebody he knew on first-name terms, who passed it on the same way.

Question 1: The letter arrived. How many hands did it pass through on the way? Write your guess, and write why you guessed that.

Part 2 — Ringville

Sixteen people live in Ringville and they sit in one circle. Everybody is friends with the two on their left and the two on their right: four friends each, thirty-two friendships in the town.



Question 2(a): Person 1 has a letter for person 9. Trace the chain on the drawing. How many handshakes does it take? ________

Question 2(b): Now everybody: in each person's box, write how many handshakes it takes to reach them from person 1. Two are filled in, and you may find you only have to work out half of the rest.

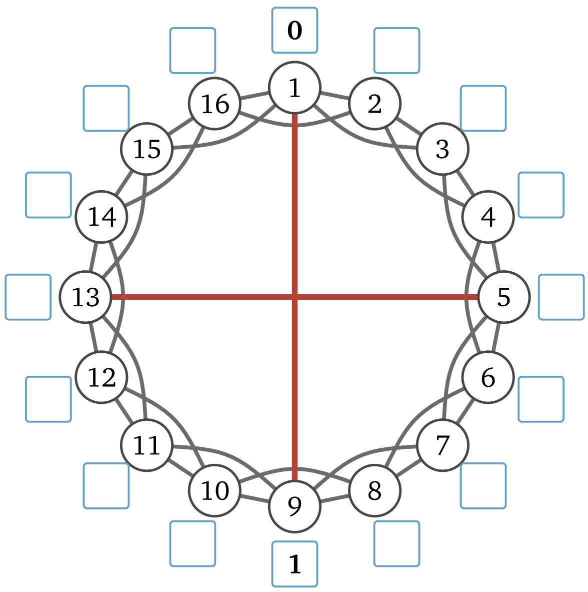
Question 2(c): Add up every number in the boxes: ________ Share that total out among the fifteen other people — the town's **average**: ________ And the biggest number in any box — the **worst case**: ________

Question 2(d): One of those two numbers you would give to a newspaper. The other you would give to an engineer who has to promise the letter arrives. Which is which, and why?

Question 3: Ringville grows. A thousand people, still one circle, still four friends each. How many handshakes from somebody to the person sitting directly opposite them? ________ And if the circle held all eight billion people on Earth? ________ So could a world wired like Ringville get the Omaha letter to Boston in six handshakes, and why?

Part 3 — Two shortcuts

Two of the sixteen went away to college and each came home with one friend from the far side of town: person 1 is now also friends with person 9, and person 5 with person 13. Nothing else changed — thirty-four friendships, not thirty-two.



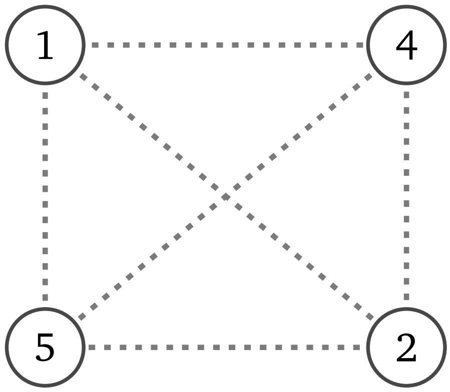
Question 4: Fill in the boxes again, for the town with the two new friendships. Two are done. Then the same three numbers: new total ________ new average ________ new worst case ________

Question 5: Thirty-two friendships became thirty-four. Work out both changes as percentages — friendships went *up* by ________% and the average went *down* by ________%. Say, in your own words, what those two lines did that thirty-two ordinary friendships could not.

Part 4 — Do your friends know each other?

Person 1 has just picked up a fifth friend, so take somebody ordinary instead. **Person 3** has four friends: persons 1, 2, 4 and 5.

Four people can be paired up six ways, and here are the six, lifted out of the circle as dashed lines.



person 3's four friends

Question 6(a): Go back to the circle on page 1 and check the six pairs one at a time. Darken the dashed lines that are real friendships. How many of the six? ________

Question 6(b): Ringville grows to ten thousand people, still four friends each. Person 3 still has four friends and they can still be paired six ways. How many of the six are friends now? ________

Question 6(c): A different town: the same sixteen people and the same thirty-two friendships, but this time who is friends with whom was drawn out of a hat. Pick two people — what is the chance they are friends? ________
So how many of person 3's six pairs would you expect to be friends? ________
And in a hat-drawn town of ten thousand, still four friends each? ________

Question 6(d): That fraction has a name — it is person 3's **local clustering**. You have now measured it four times. Say in one sentence what makes it big and what makes it small.

Three people who are all friends with each other make a **triangle**. Ringville, before the two shortcuts, has sixteen of them.

Question 7(a): Then the two shortcuts went in. Do persons 1 and 9 have a friend in common? Do 5 and 13? So how many triangles now? ________

Question 7(b): The real model does not *add* a shortcut, it **moves** one: person 1 drops person 2 and befriends person 9 instead. Which friends did persons 1 and 2 have in common? ________ So how many triangles does the town lose? ________

Question 7(c): Put Question 5 beside 7(a) and 7(b). What does a shortcut buy the town, and what does it cost?

Part 5 — Back to Omaha

Question 8(a): Everybody knows 150 people. Pretend for a moment that no two of your friends know each other, so every handshake reaches 150 people nobody in the chain has met yet. Round as you like.

handshakes	people reached
1	150
2	22,500
3	about 3.4 million
4	
5	

At how many handshakes do you pass eight billion? ________

Compare that with your guess in Question 1.

Question 8(b): Question 6 says the pretending is false. Which way does that push the answer — more handshakes, or fewer? Why?

Question 8(c): You are person 3 in the town with the two shortcuts, holding a letter for person 12. You cannot see the drawing: you know only your own four friends and the chair each sits in. Hand it to whichever of them sits closest to chair 12; they know only *their* friends and do the same. Play it out one person at a time.

The chain: ____________________________________ handshakes: ________

Now look at the whole drawing. What is the shortest chain from person 3 to person 12, and how long is it? ____________________

Question 8(d): Of 296 letters sent from Omaha, 64 arrived, in five or six steps. Is "a short chain exists" the same claim as "an ordinary person can find one"? Which did the letters test, and what do the 232 that never arrived say about it?

Part 6 — The lab, on your own

Do this one by yourself, with this sheet beside you. The notebook builds the same town and counts its handshakes and its triangles for you — once you have written the three rules it runs on.

Point a camera at the square. If that fails, type `go.skojaku.com/m02lab`